

CM0845 Logic

Propositional Logic: Some Properties

Andrés Sicard-Ramírez

Universidad EAFIT

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Preliminaries

Textbook

Dirk van Dalen (2013). Logic and Structure. Section 2.3.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Finite Conjunctions and Disjunctions

Definition 2.3.8

Finite conjunctions and **finite disjunctions** are inductively defined.

$$\bigwedge_{i \leq 0} \varphi_i = \varphi_0,$$
$$\bigwedge_{i \leq n+1} \varphi_i = \bigwedge_{i \leq n} \varphi_i \wedge \varphi_{n+1}.$$

$$\bigvee_{i \leq 0} \varphi_i = \varphi_0,$$
$$\bigvee_{i \leq n+1} \varphi_i = \bigvee_{i \leq n} \varphi_i \vee \varphi_{n+1}.$$

References

Dirk van Dalen [1980] (2013). Logic and Structure. 5th ed. Springer (cit. on p. 2).